

First Order Linear ODEs (Exact/Non-exact)

- *Linear differential equation of order ONE:*

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Example.

$$2(y - 4x^2)dx + 2xdy = 0 \Rightarrow \frac{dy}{dx} + \frac{1}{x}y = 4x \quad [\text{Exact ODE}]$$

$$2(y - 4x^2)dx + xdy = 0 \Rightarrow \frac{dy}{dx} + \frac{2}{x}y = 8x \quad [\text{Non-exact ODE}]$$

$$2xy \, dx + dy = 0 \Rightarrow \frac{dy}{dx} + 2xy = 0 \quad [\text{Non-exact, separable ODE}]$$

$$(y - x)dx + dy = 0 \Rightarrow \frac{dy}{dx} + y = x \quad [\text{Non-exact ODE}]$$

- ❖ In some instances a first order linear ODEs (non-exact) can be solved by separation of variables.

First Order Linear ODEs (Exact/Non-exact)

Steps to solve a First order linear ODEs (non-exact and non-separable):

- (i) Put the equation into the standard form: $\frac{dy}{dx} + P(x)y = Q(x)$.
 - (ii) Obtain the integrating factor with no integral constant adding: $f(x) = e^{\int P(x)dx}$.
 - (iii) Multiply the both sides of the standard form equation by the integrating factor.
- The resultant equation will be then exact ODEs.
- (iv) Solve the resultant exact ODEs.

Linear First Order Non-Homogeneous ODE: $\frac{dy}{dx} + P(x)y = Q(x)$.

Linear First Order Homogeneous ODE: $\frac{dy}{dx} + P(x)y = 0$.

First Order Linear ODEs (Exact/Non-exact)

Example. Solve the differential equations,

$$2(y - 4x^2)dx + xdy = 0$$

[The ODE is not exact]

Solution. The given equation is not exact since

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(2y - 8x^2) = 2, \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(x) = 1$$

Now, rewrite the given equation into the standard form as

$$\frac{dy}{dx} + \frac{2(y - 4x^2)}{x} = 0 \Rightarrow \frac{dy}{dx} + \frac{2}{x}y = 8x \quad \text{when } x \neq 0$$

Then the integrating factor yields,

$$e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2$$

Now multiplying the integrating factor in the both side of the given ODE yields,

$$x^2 \frac{dy}{dx} + 2xy = 8x^3 \Rightarrow x^2 dy + 2xy dx = 8x^3 dx \Rightarrow (x^2) d\textcolor{violet}{y} + \textcolor{violet}{y} d(x^2) = 8x^3 dx$$

$$\Rightarrow d(x^2 y) = 8x^3 dx \Rightarrow \int d(x^2 y) = \int 8x^3 dx$$

$$d(f \cdot g) = f \cdot dg + g \cdot df$$

$$\therefore x^2 y = 2x^4 + c$$

First Order Linear ODEs (Exact/Non-exact)

Example. Solve the differential equations,

$$\frac{dy}{dx} - 3y = 6$$

[The ODE is not separable]

Solution. Here, the integrating factor becomes,

$$e^{\int (-3)dx} = e^{-3x}$$

Now multiplying the integrating factor in the both side of the given ODE yields,

$$e^{-3x} \frac{dy}{dx} - 3e^{-3x}y = 6e^{-3x} \Rightarrow e^{-3x}dy - 3e^{-3x}ydx = 6e^{-3x}dx$$

$$\Rightarrow (e^{-3x})d\mathbf{y} + \mathbf{y} d(e^{-3x}) = 6e^{-3x}dx$$

$$\Rightarrow d(\mathbf{y}e^{-3x}) = 6e^{-3x}dx$$

$$d(f \cdot g) = f \cdot dg + g \cdot df$$

$$\Rightarrow \int d(ye^{-3x}) = 6 \int e^{-3x} dx$$

Therefore, the desired solution becomes,

$$\therefore \mathbf{y}e^{-3x} = -\mathbf{2}e^{-3x} + \mathbf{c} \Rightarrow \mathbf{y} = \mathbf{c}e^{3x} - \mathbf{2}$$

First Order Linear ODEs (Exact/Non-exact)

Example. Solve the initial value problem,

$$\frac{dy}{dx} + y = x, \quad y(0) = 4 \quad [\text{The ODE is not separable}]$$

Solution. Here, the integrating factor becomes,

$$e^{\int (1)dx} = e^x$$

Now multiplying the integrating factor in the both side of the given ODE yields,

$$e^x \frac{dy}{dx} + e^x y = xe^x \Rightarrow e^x dy + e^x y dx = xe^x dx \Rightarrow (e^x) d(\textcolor{violet}{y}) + \textcolor{violet}{y} d(e^x) = xe^x dx$$

$$\Rightarrow d(\textcolor{violet}{y}e^x) = xe^x dx \Rightarrow \int d(\textcolor{violet}{y}e^x) = \int xe^x dx$$

$$\therefore \textcolor{violet}{y}e^x = xe^x - e^x + c \Rightarrow \textcolor{violet}{y} = x - 1 + ce^{-x}$$

$$d(f \cdot g) = f \cdot dg + g \cdot df$$

From the initial condition, we have, when $x = 0, y = 4$. Therefore, $4 = 0 - 1 + c \Rightarrow c = 5$.

Therefore the particular solution of the problem is, $\textcolor{violet}{y} = x - 1 + 5e^{-x}$

Here, ce^{-x} in the general solution is called a transient term since $e^{-x} \rightarrow 0$ as $x \rightarrow \infty$.

First Order Linear ODEs (Exact/Non-exact)

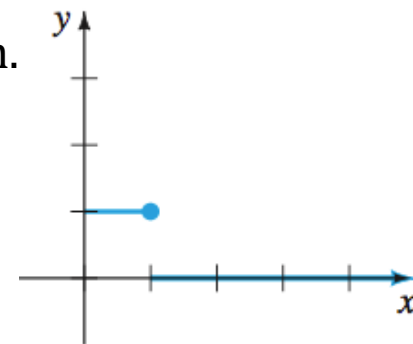
Example. Solve the initial value problem,

$$\frac{dy}{dx} + y = f(x), \quad y(0) = 0 \quad \text{where} \quad f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$$

Solution. Here, $f(x)$ is discontinuous at $x = 1$ as can be seen in the graph.

Here, the integrating factor becomes,

$$e^{\int (1)dx} = e^x$$



Now multiplying the integrating factor in the both side of the given ODE yields,

$$\begin{aligned} e^x \frac{dy}{dx} + e^x y &= f(x)e^x \Rightarrow e^x dy + e^x y dx = f(x)e^x dx \\ &\Rightarrow (e^x)dy + y d(e^x) = f(x)e^x dx \\ &\Rightarrow d(ye^x) = f(x)e^x dx \\ &\Rightarrow \int d(ye^x) = \int f(x)e^x dx \end{aligned}$$

First Order Linear ODEs (Exact/Non-exact)

Example. Solve the initial value problem,

$$\frac{dy}{dx} + y = f(x), \quad y(0) = 0 \quad \text{where} \quad f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$$

Solution. Continued...

$$ye^x = \begin{cases} \int e^x dx, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases} = \begin{cases} e^x + c_1, & 0 \leq x \leq 1 \\ c_2, & x > 1 \end{cases} \Rightarrow y = \begin{cases} 1 + c_1 e^{-x}, & 0 \leq x \leq 1 \\ c_2 e^{-x}, & x > 1 \end{cases}$$

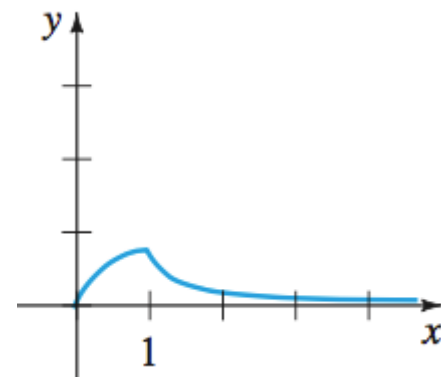
From the initial condition, we have, when $x = 0$, $y = 0$. Therefore, we must have $c_1 = -1$.

For the continuity of $y(x)$ at any point, it is required that $y(x)$ must be continuous at $x = 1$ and then we must have

$$\lim_{x \rightarrow 1^+} y(x) = y(1) \Rightarrow \lim_{x \rightarrow 1^+} c_2 e^{-x} = 1 + c_1 e^{-1} \Rightarrow c_2 e^{-1} = 1 - e^{-1} \Rightarrow c_2 = e - 1.$$

Thus the desired solution yields,

$$y = \begin{cases} 1 - e^{-x}, & 0 \leq x \leq 1 \\ (e - 1)e^{-x}, & x > 1 \end{cases}$$



First Order Linear ODEs (Exact/Non-exact)

Example. Solve the differential equations,

$$(y + 1)dx + (4x - y)dy = 0 \quad [\text{The ODE is not linear in } y, \text{ but in } x]$$

Solution. The given equation is not linear in y , but it is linear in x . Rewriting the equation,

$$\frac{dx}{dy} + \frac{4x - y}{y + 1} = 0 \Rightarrow \frac{dx}{dy} + \frac{4}{y + 1}x = \frac{y}{y + 1} = 1 - \frac{1}{y + 1}$$

Here, the integrating factor becomes,

$$e^{\int \left(\frac{4}{y+1}\right) dy} = e^{4 \ln(y+1)} = e^{\ln(y+1)^4} = (y + 1)^4$$

Now multiplying the integrating factor in the both side of the ODE yields,

$$(y + 1)^4 dx + 4(y + 1)^3 x dy = [(y + 1)^4 - (y + 1)^3] dy$$

$$\Rightarrow (y + 1)^4 dx + x d\{(y + 1)^4\} = [(y + 1)^4 - (y + 1)^3] dy$$

$$\Rightarrow d[x(y + 1)^4] = [(y + 1)^4 - (y + 1)^3] dy$$

$$d(f \cdot g) = f \cdot dg + g \cdot df$$

$$\Rightarrow \int d[x(y + 1)^4] = \int [(y + 1)^4 - (y + 1)^3] dy \Rightarrow x(y + 1)^4 = \frac{(y + 1)^5}{5} - \frac{(y + 1)^4}{4} + \frac{c}{20}$$

$$\Rightarrow x = \frac{y + 1}{5} - \frac{1}{4} + \frac{c}{20} (y + 1)^{-4} \Rightarrow \mathbf{20x = 4y - 1 + c(y + 1)^{-4}}$$

Reduction to First Order Linear ODEs

Bernoulli's Equation:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad \text{where } n \in \mathbb{R}$$

For $n = 0, 1$, the above equation becomes linear.

For $n \neq 0, 1$, the above equation becomes non-linear and can be reduced into linear by substituting $u = y^{1-n}$

Example. Solve the following Bernoulli differential equation,

$$x \frac{dy}{dx} + y = x^2 y^2$$

Solution. The given equation is not linear in y . Rewriting the equation in the form,

$$\frac{dy}{dx} + \frac{1}{x}y = xy^2 \Rightarrow y^{-2} \frac{dy}{dx} + \frac{1}{x}y^{-1} = x$$

Now substitute, $u = y^{-1} \Rightarrow \frac{du}{dx} = -y^{-2} \frac{dy}{dx} \Rightarrow y^{-2} \frac{dy}{dx} = -\frac{du}{dx}$. Thus the ODE becomes,

$$y^{-2} \frac{dy}{dx} + \frac{1}{x}y^{-1} = x \Rightarrow -\frac{du}{dx} + \frac{1}{x}u = x \Rightarrow \frac{du}{dx} - \frac{1}{x}u = -x \quad \text{[The ODE is linear in } u \text{]}$$

Reduction to First Order Linear ODEs

Example. Solve the following Bernoulli differential equation,

$$x \frac{dy}{dx} + y = x^2 y^2$$

Solution. Continued...

Here, the integrating factor becomes, $e^{\int (-\frac{1}{x}) dx} = e^{-\ln x} = e^{\ln x^{-1}} = x^{-1} = \frac{1}{x}$

Now multiplying the integrating factor in the both side of the ODE yields,

$$\frac{1}{x} \frac{du}{dx} - \frac{1}{x^2} u = -\frac{x}{x} \Rightarrow \frac{du}{dx} - \frac{u}{x} = -x \Rightarrow x du - u dx = -x^2 dx$$

$$\Rightarrow \frac{x du - u dx}{x^2} = -\frac{x^2 dx}{x^2}$$

$$d\left(\frac{f}{g}\right) = \frac{g \cdot df - f \cdot dg}{g^2}$$

$$\Rightarrow d\left(\frac{u}{x}\right) = -dx \Rightarrow \int d\left(\frac{u}{x}\right) = -\int dx$$

$$\Rightarrow \frac{u}{x} = -x + c \Rightarrow u = -x^2 + cx \Rightarrow \frac{1}{y} = -x^2 + cx$$

The desired solution becomes, $y = \frac{1}{cx - x^2}$

First Order Linear ODEs (Exact/Non-exact)

Exercises 2.3

H.W. from the text book

Find the general solution of the given differential equation. Determine whether there are any transient terms in the general solution.

1. $\frac{dy}{dx} = 5y$

2. $\frac{dy}{dx} + 2y = 0$

3. $\frac{dy}{dx} + y = e^{3x}$

4. $3\frac{dy}{dx} + 12y = 4$

5. $y' + 3x^2y = x^2$

6. $y' + 2xy = x^3$

7. $x^2y' + xy = 1$

8. $y' = 2y + x^2 + 5$

9. $x\frac{dy}{dx} - y = x^2 \sin x$

10. $x\frac{dy}{dx} + 2y = 3$

11. $x\frac{dy}{dx} + 4y = x^3 - x$

12. $(1+x)\frac{dy}{dx} - xy = x + x^2$

13. $x^2y' + x(x+2)y = e^x$

16. $y dx = (ye^y - 2x) dy$

14. $xy' + (1+x)y = e^{-x} \sin 2x$

17. $\cos x \frac{dy}{dx} + (\sin x)y = 1$

15. $y dx - 4(x + y^6) dy = 0$

Solve the given initial-value problem. Give the largest interval over which the solution is defined

25. $\frac{dy}{dx} = x + 5y, \quad y(0) = 3$

26. $\frac{dy}{dx} = 2x - 3y, \quad y(0) = \frac{1}{3}$

27. $xy' + y = e^x, \quad y(1) = 2$

28. $y\frac{dx}{dy} - x = 2y^2, \quad y(1) = 5$

31. $x\frac{dy}{dx} + y = 4x + 1, \quad y(1) = 8$

32. $y' + 4xy = x^3e^{x^2}, \quad y(0) = -1$

33. $(x+1)\frac{dy}{dx} + y = \ln x, \quad y(1) = 10$

34. $x(x+1)\frac{dy}{dx} + xy = 1, \quad y(e) = 1$

35. $y' - (\sin x)y = 2 \sin x, \quad y(\pi/2) = 1$

36. $y' + (\tan x)y = \cos^2 x, \quad y(0) = -1$

First Order ODEs - Exact equations

Exercises 2.3

H.W. from the text book

Solve the given initial-value problem.

37. $\frac{dy}{dx} + 2y = f(x)$, $y(0) = 0$, where

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 3 \\ 0, & x > 3 \end{cases}$$

39. $\frac{dy}{dx} + 2xy = f(x)$, $y(0) = 2$, where

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$$

38. $\frac{dy}{dx} + y = f(x)$, $y(0) = 1$, where

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ -1, & x > 1 \end{cases}$$

40. $(1 + x^2) \frac{dy}{dx} + 2xy = f(x)$, $y(0) = 0$, where

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ -x, & x \geq 1 \end{cases}$$

53. A heart pacemaker consists of a switch, a battery of constant voltage E_0 , a capacitor with constant capacitance C_0 , and the heart as a resistor with constant resistance R_0 . When the switch is closed, the capacitor charges; when the switch is open, the capacitor discharges, sending an electrical stimulus to the heart. During the time the heart is being stimulated, the voltage E across the heart satisfies the linear differential equation $\frac{dE}{dt} = -\frac{1}{R_0 C_0} E$.

Solve the DE, subject to $E(4) = E_0$

First Order ODEs – Bernoulli's equations

Exercises 2.5

H.W. from the text book

Solve the following Bernoulli's differential equation by using an appropriate substitution.

15. $x \frac{dy}{dx} + y = \frac{1}{y^2}$

16. $\frac{dy}{dx} - y = e^x y^2$

21. $x^2 \frac{dy}{dx} - 2xy = 3y^4, \quad y(1) = \frac{1}{2}$

17. $\frac{dy}{dx} = y(xy^3 - 1)$

18. $x \frac{dy}{dx} - (1 + x)y = xy^2$

22. $y^{1/2} \frac{dy}{dx} + y^{3/2} = 1, \quad y(0) = 4$

38. In the study of population dynamics one of the most famous models for a growing but bounded population is the **logistic equation**

$$\frac{dP}{dt} = P(a - bP)$$

where a and b are positive constants. Solve the DE using the fact that it is a Bernoulli equation.